

Teardown #01 — What Does an AI-Server Accelerator Actually Cost to Build?

Draft for the first public silicon-economics teardown post. All figures are modeled estimates from public and analyst-published data.

DRAFT v0.1 for founder review. Publication targets: company blog, LinkedIn, HN. Embed the interactive demo Build via share link before publishing.

Standfirst

AI accelerator pricing is public. Cost structure is not. Below we build one, deterministically: a 4-chiplet, N3-class, CoWoS-packaged AI-server accelerator, modeled end to end with every assumption sourced, dated, and confidence-rated — and every output traceable to a named formula. No black boxes; you can open the live model and change any number.

The Build

- 4 × 145 mm² compute chiplets, N3-class node (analyst-estimate wafer price: \$18,000; defect density 0.12/cm²).
- 1 × I/O die on a mature node.
- CoWoS-S silicon interposer, ~760 mm², modeled at \$3.50/mm² with 96% assembly yield.
- HBM stacks excluded from this teardown (memory economics deserve their own issue — subscribe).

Step 1 — Silicon: why chiplets win the yield war

Murphy's yield model at N3-class defect density gives a 145 mm² chiplet roughly 84% yield. A hypothetical monolithic ~600 mm² version of the same design yields dramatically worse — this is the entire economic argument for disaggregation. But chiplets must ALL be good: four compute dies plus I/O stacked multiplicatively (known-good-die flow) put compound silicon yield in the 40–50% band before packaging. Disaggregation buys per-die yield and spends some of it back on compounding.

Step 2 — The punchline: the interposer out-costs the silicon

A ~760 mm² CoWoS-S interposer at \$3.50/mm², divided by its 96% assembly yield, models to roughly \$2,770 per unit. That is more than all five silicon dies combined. In the advanced-packaging era, the package is the product. This single line item is why CoWoS allocation — not wafer supply — is the binding constraint on AI accelerator volume.

Step 3 — The unit stack (modeled estimates)

Cost element	Modeled \$/unit (approx.)	Notes
Compute + I/O silicon (KGD-adjusted)	~\$400–500	Murphy yield per die; compound KGD flow
CoWoS-S interposer (yield-adjusted)	~\$2,770	760 mm ² × \$3.50/mm ² ÷ 0.96
Assembly, substrate, final test	~\$70–100	Incl. test seconds × cost/second
Total unit COGS	~\$3,200	Before NRE amortization

Step 4 — Program economics

- At an \$8,500 ASP (H100/MI300-class market), gross margin models to ~62%.

- NRE: ~\$260M (masks, IP, design + verification labor) !' break-even near 5 production at AI-demand run rates.
- The time dimension matters more than the point estimate: defect-density learning curves, ASP erosion, and a 35%-probability respin scenario (mask set + 2 quarters) all shift the answer. The interactive model computes all three.

What this means

- If you are choosing between monolithic and chiplets at advanced nodes, the decision is a packaging-economics decision.
- Supply-constrained programs should model CoWoS allocation as the demand ceiling, not wafer starts.
- Any cost model without yield ramp and respin probability is a snapshot, not a plan.

Open the model

Every number above comes from a deterministic Build you can open, inspect, and edit in your browser — no sign-in, nothing transmitted: [DEMO SHARE LINK]. Click any metric for its formula, inputs, assumptions, and data vintage.

Sources & confidence

Wafer pricing, defect density, and CoWoS cost assumptions are analyst estimates and published disclosures (medium confidence, dated in-model). No NDA-covered foundry pricing is used — by policy. Methodology (Murphy yield, KGD semantics, packaging cost model) is documented openly in the app's formula library.